Plane mirrors

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the reflection law and locate a virtual image.

SUCCESS CRITERIA - I can...

1. Measure angles from the normal.
2. Use equal incidence and reflection angles.
3. Place the image the same perpendicular distance behind the mirror.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Sensors 1.02 m apart record delay 0.003 s. Find wave speed. If 0.0005 s is added to the delay, find the incorrect speed and describe the bias.

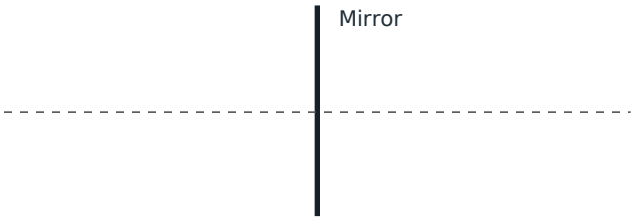
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\theta_{incident} = \theta_{reflected}$
image distance behind = object distance in front.
Plane mirror; measure angles from the normal. The image is virtual.

Worked reasoning: An object 2 m in front of an ideal plane mirror has a virtual image 2 m behind it. The object-image separation is 4 m; reflected rays appear to come from the image.

Your turn: Draw the mirror, object, image and backward extensions of two reflected rays.

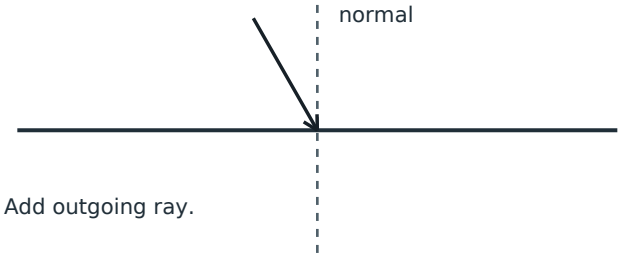


Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw the normal at the mirror. Complete a reflected ray for an incident ray 30° from normal.

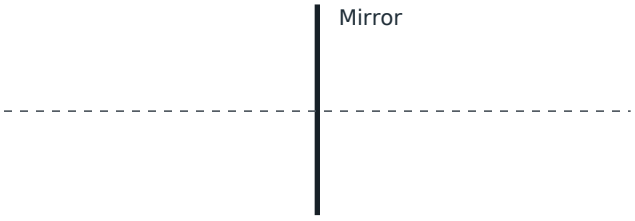


Add outgoing ray.

Marzano check - circle: ① ② ③

04 Apply the physics

A laser ray makes a 20° angle with a plane mirror's surface. Draw the normal and both rays. Find incidence angle measured from the normal, then the angle between the incident and reflected ray segments drawn away from the contact point.



Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

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Continue down each column.

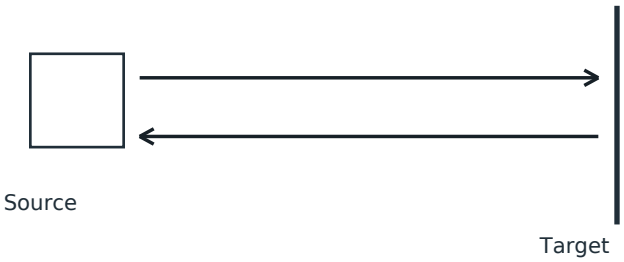
05 Explain your reasoning

Why is the mirror image called virtual rather than a real object behind the glass?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

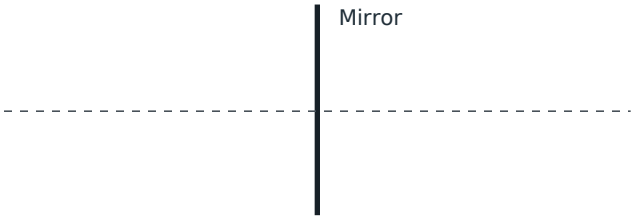
Echoes return at 0.004 s and 0.006 s with the same wave speed. Compare ranges. Does delay alone determine mass?
Label the outgoing and returning paths.



Marzano check - circle: 1 2 3

07 Exit check

An object moves 0.50 m toward a fixed mirror. Find image movement and change in object-image separation.

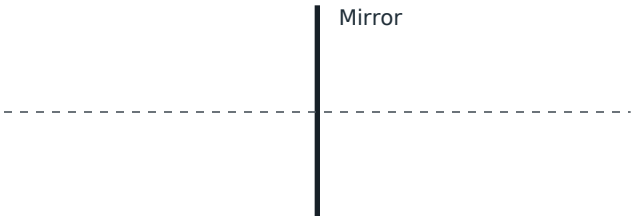


Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

An object moves toward a fixed plane mirror at 1.2 m/s. Find image velocity relative to the mirror and the rate at which object-image separation decreases.



Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3